Content-Based Image Retrieval Systems: A Survey

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1 Introduction

In many areas of commerce, government, academia, and hospitals, large collections of digital images are being created. Many of these collections are the product of digitizing existing collections of analogue photographs, diagrams, drawings, paintings, and prints. Usually, the only way of searching these collections was by keyword indexing, or simply by browsing. Digital images databases however, open the way to content-based searching. In this paper we survey some technical aspects of current content-based image retrieval systems.

A number of other overviews on image database systems, image retrieval, or multimedia information systems have been published, see e.g. [60], [20], [22], [25], [21], [15], [47], [7]. The purpose of *this* survey however, is to provide an overview of the functionality of contemporary image retrieval systems in terms of technical aspects: querying, relevance feedback, features, matching measures, indexing data structures, and result presentation. It compares specific systems, rather than general architectures, and provides a basis for (or defense against) statements like "this or that system already does what your system does". It also is a thorough foundation for claims that most systems use low level features, and few use high level, semantically meaningful features.

A number of keyword-based general WWW search engines allow to indicate that the media type must be images. HotBot (`http://hotbot.lycos.com/`), and NBCi (`http://www.nbci.com/`) are examples of these. A number of other general search engines are more specifically for images, such as Yahoo!'s Picture Gallery (`http://gallery.yahoo.com/`) or the multimedia searcher of Lycos (`http://multimedia.lycos.com/`), but they are still only keyword based. There are many special image collections on the web that can be searched with a number of alphanumerical keys. For example, ImageFinder (`http://sunsite.berkeley.edu/ImageFinder/`) provides a list of such collections as a tool to help teachers locate historical photographs from collections around the world.

AltaVista Photofinder is a search engine that allows content-based image retrieval, both from special collections, and from the Web. In the remainder of this paper, we will give an overview of other such content-based image retrieval systems, both commercial/production systems, and research/demonstration systems.

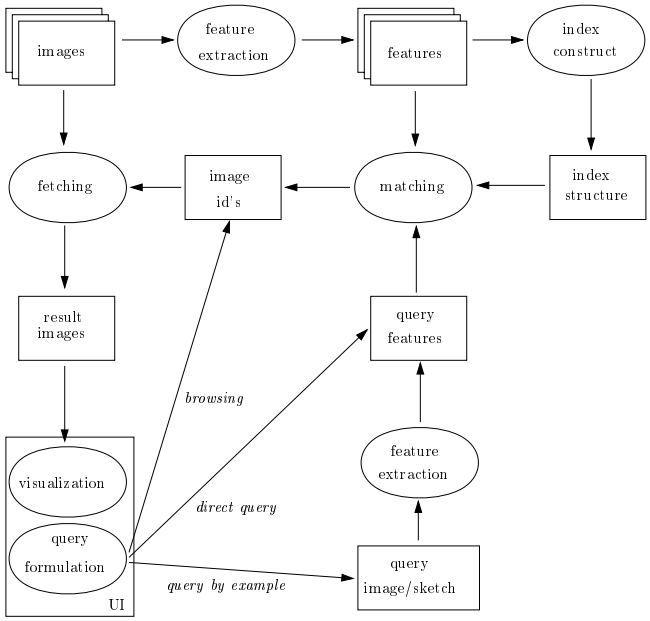


Figure 1: Content-based image retrieval framework.

2 Criteria

Many image retrieval systems can be conceptually described by the framework depicted in figure 1. In this article we survey how the user can formulate a query, whether and how relevance feedback is possible, what kind of features are used, how features from query image and data base image are matched, how the retrieval results are presented to the user, and what indexing data structures are used.

The user interface typically consists of a query formulation part and a result presentation part. Specification of which images to retrieve from the database can be done in many ways. One way is to browse through the database one by one. Another way is to specify the image in terms of keywords, or in terms of image features that are extracted from the image, such as a color histogram. Yet another way is to provide an image or sketch from which features of the same type must be extracted as for the database images, in order to match these features. A nice taxonomy of interaction models is given in [63]. Relevance feedback is about providing positive or negative feedback about the retrieval result, so that the system can refine the search.

We will consider several classes of features that are used to specify queries: color, texture, shape, spatial layout, and faces. Color features are often easily obtained directly from the pixel intensities, e.g. color histogram over the whole image, over a fixed subimage, or over a segmented region are often used. Although a precise definition of texture is untraceable, the notion of texture generally refers to the presence of a spatial pattern that has some properties of homogeneity. In particular, the homogeneity cannot result from the presence of only a single color in the regions, but requires interaction of various colors.

There is no universal definition of what shape is either. Impressions of shape can be conveyed by color

or intensity patterns, or texture, from which a geometrical representation can be derived. This is shown already in Plato's work Meno [46]. (This is one of the so-called Socratic dialogs, where two persons discuss aspects of virtue; to memorialize Socrates, one of the figures is called after him.) In this work, the word 'figure' is used for shape. Socrates' description "figure is the only existing thing that is found always following color" does not satisfy Meno, after which Socrates gives a definition in "terms employed in geometrical problems": "figure is limit of solid". In this paper too we consider shape as something geometrical. Therefore, we consider for example edge orientation over all pixels as texture, but edge orientation at only region contours as shape information. Shape descriptors are diverse, e.g. turning angle functions, deformable templates, algebraic moments, and Fourier coefficients. For an overview of shape matching techniques, see [61].

The term indexing is often used for identifying features within an image; with indexing data structures we here mean structures to speed up the retrieval of features, and thus the images containing them, within large image collections.

Spatial layout is about the absolute or relative position of color, texture, or shape information within the image. Higher level features are increasingly more specific, and thus less widely used. However, faces are frequently present in pictures and relatively often used as a feature, so that we tally its use separately.

3 Anthology of Systems

We have surveyed 50 content-based image retrieval systems in terms of querying, relevance feedback, features, matching measures, indexing data structures, and result presentation. Below we describe only a number of these, but in the end we give a summarizing table containing them all. In [62] the full description of all systems is available.

The choice of these seven is somewhat arbitrary, but they all show some particularly strong element: Blobworld its segmentation and focus on objects, ImageMiner its high semantic level, PicHunter its relevant feedback, PicToSeek its invariant color features, SMURF its indexing method and shape matching, SYNAPSE its appearance approach, VisualSEEk its spatial layout, WISE its wavelet based texture features.

If no 'querying', 'features', 'matching', 'indexing', 'relevance feedback', or 'result presentation' is mentioned, than it is not applicable to the system (e.g. there is no relevance feedback), or no such information is known to us (e.g. often no information is given about indexing data structures).

3.1 Blobworld

Developer Computer Science Division, University of California, Berkeley.

URL `http://elib.cs.berkeley.edu/photos/blobworld/`. A demo of Blobworld is available there too.

References [9].

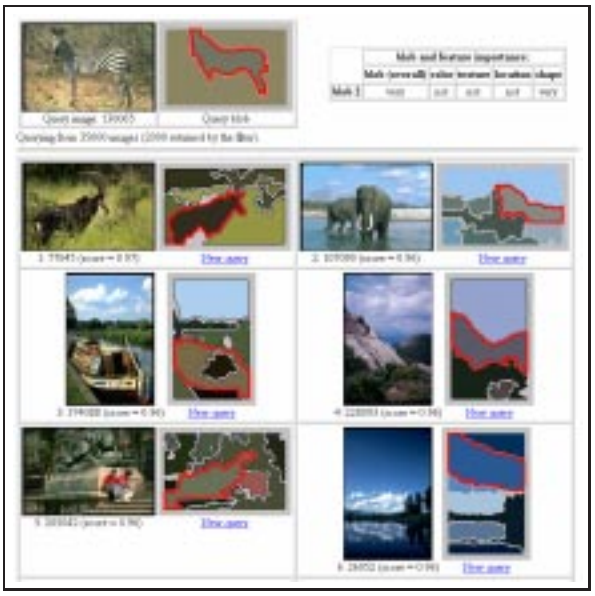


Figure 2: Blobworld query result.

Features The features used for querying are the color, texture, location, and shape of regions (blobs) and of the background. The color is described by a histogram of 218 bins of the color coordinates in Lab-space. Texture is represented by mean contrast and anisotropy over the region, as the 2D coordinate $(contrast, contrast \times anisotropy)$. Shape is represented by (approximate) area, eccentricity, and orientation.

Querying The user first selects a category, which already limits the search space. In an initial image, the user selects a region (blob), and indicates the importance of the blob ('somewhat', 'very'). Next, the user indicates the importance of the blob's color, texture, location, and shape ('not', 'somewhat', 'very'). More than one regions can be used for querying.

Matching To match two color histograms h_1 and h_2, the quadratic form distance is used: $d(h_1, h_2) = (h_1 - h_2)^T A (h_1 - h_2)$, where $A = (a_{ij})$ is a symmetric matrix of weights representing the similarity between color bins i and j. The distance between two texture descriptors is the Euclidean distance between their coordinates in representation space. The distance between centroids is the Euclidean distance. The distances are combined into a single final distance.

Indexing Rather than actually computing the distances between the full color histogram vectors of length 218 as $d(h_1, h_2) = (h_1 - h_2)^T A (h_1 - h_2)$, singular value decomposition (SVD) is used to project the histogram vectors onto a lower-dimensional subspace. The resulting points are indexed by an R*-tree [3].

Result presentation The retrieved images are ranked in linear order, and presented together with the segmented version showing the regions, see figure 2.

Applications The demo on the web provides retrieval from a collection of 10000 Corel stock photos.

3.2 ImageMiner

Developer Technologie-Zentrum Informatik, University of Bremen, Germany.

URL http://www.tzi.de/bv/ImageMinerhtml/.

References [28].

Features ImageMiner generates content descriptions of images based on color, texture and contours. A color segmentation of the image is done by first computing a color histogram for all grid elements of homogeneous size that the image is divided into and then grouping the grid elements according to their most frequent color in the histogram. Bounding rectangles are determined for the grouped grids. For each rectangle, attributes like size, position (both relative to the grid size), associated color and color density (the ratio of the size of color rectangle to the amount of grid elements with that color) are stored.

The texture segmentation is similarly given by the bounding rectangles of the grids grouped according to their similar texture. For representing texture, a statistical approach is adopted. For every grid element, the cooccurrence matrix P is calculated for four orientations ($0°$, $45°$, $90°$ and $135°$). This is an $N_g \times N_g$ matrix (with N_g representing the number of distinct gray levels in the quantized image) where P_{ij} is defined as the joint probability that a pair of neighboring pixels, one with gray-level i and the other with gray-level j, occur in the image under the four specific angular relationships. From each matrix, five statistical features are determined (angular second moment, contrast, correlation, variance and entropy). The average values of these features over the four orientations are the inputs of a neural network trained with a backpropagation algorithm. The network output neurons symbolize seven visual texture properties: bloblikeness, multiareas, planarity, coarseness, regularity, directionality and softness. For every texture rectangle determined, its size, position, the classified texture and the texture density are stored.

The contour-based shape description is obtained by first detecting the edge points using a gradient method and then connecting these points to form contours. For the closed regions detected some shape parameters are extracted (the coordinate of the middle point, the size and the bound coordinates).

The neighborhood relations of the contours and the color and texture regions are represented in a graph. Graph parsing algorithms classify scenes and generate a description with keywords and values.

Querying SQL-like queries with keywords and values can be posed.

Matching and Indexing Retrieval is performed by the IBM Search Manager text searcher.

Applications Videos are made available for retrieval with ImageMiner in two steps. First, shots are extracted from the video by using a histogram based method. More precisely, a new shot is detected when the difference in the intensity histogram of two consecutive frames exceeds a given threshold. Then, a mosaicing-technique is used to generate a single still image for each shot detected in the video sequence. This mosaiced image comprises all the information in a shot and is analyzed by the ImageMiner system as above.

3.3 PicHunter

Developer NEC Research Institute, Princeton, NJ, USA.

References [13].

Features The current version of the systems uses color histogram and color spatial distribution along with hidden textual annotations. Besides a 64-bin HSV histogram, two other vectors – a 256-length HSV color autocorrellogram (CORR) and a 128-length RGB color-coherence vector (CCV) – are describing the color content of an image. The first 64 components of CORR represent the number of pixels of a particular color from the 64-quantized HSV color space having neighbors of the same color at distance 1. The rest of the vector is defined in the same way for distances 3, 5 and 7. CCV is made of 64 coherence pairs, each pair giving the number of coherent, and incoherent pixels of a particular discretized color in the RGB space. In order to classify pixels in one of the two categories, the image is first blurred slightly by replacing each pixel value with the average value of the pixels in its 3×3 neighborhood. Then, pixels are grouped in connected components. A pixel will be classified as coherent if the connected component it belongs to is larger than a fixed threshold. Keywords, selected from a set of 138 words, are associated with each image in the database population step and are represented as a boolean vector.

Querying Retrieval is done with query by example.

Matching The distance between individual features (color vectors or annotation lists represented as binary vectors) is the L_1 distance. These distances are scaled and combined in a global distance. The scaling factors are computed by maximizing the probability of a training set.

Relevance feedback PicHunter implements a probabilistic relevance feedback mechanism, which tries to predict the target image the user wants based on his actions (the images he selects as similar to the target in each iteration of a query session). A vector is used for retaining each image's probability of being the target. This vector is updated at each iteration of the relevance feedback, based on the history of the session (images displayed by the system and user's actions in previous iterations). The updating formula is based on Bayes' rule. If the n database images are noted T_j, $j = 1, \ldots, n$, and the history of the session through iteration t is denoted $H_t = \{D_1, A_1, D_2, A_2, \ldots, D_t, A_t\}$, with D_j and A_j being the images displayed by the system and, respectively, the action taken by the user at the iteration j, then the iterative update of the probability estimate of an image T_i being the target, given the history H_t, is:

$$P(T = T_i|H_t) = P(T = T_i|D_t, A_t, H_{t-1}) = \frac{P(A_t|T = T_i, D_t, H_{t-1})P(T = T_i|H_{t-1})}{\sum_{j=1}^{n} P(A_t|T = T_j, D_t, H_{t-1})P(T = T_j|H_{t-1})}.$$

In computing the probability of a user to take a certain action A_t given the history so far and the fact that the target is indeed T_i, namely $P(A_t|T = T_i, D_t, H_{t-1})$, a few models were tested. One approach is to estimate the probability of the user to pick an image X_a from $X_1, \ldots, X_{n_t}$ by

$$p_{softmin}(A = a|X_1, \ldots, X_{n_t}, T) = \frac{exp(-d(X_a, T)/\sigma)}{\sum_{i=1}^{n_t} exp(-d(X_i, T)/\sigma)},$$

and, in the case of choosing any number of images, to assume that each image is selected independently according to a $p_{softmin}$.

Result presentation While older versions were displaying the n_t images with the highest probability of being the target, in the newer version the images selected for display are determined by minimizing the expected number of future iterations estimated by entropy.

Applications The system was tested with a database gathering images from 45 CD's of Corel stock photographs.

3.4 PicToSeek

Developer Department of Computer Science, University of Amsterdam, The Netherlands.

URL http://www.science.uva.nl/research/isis/pictoseek/. A demo of the system is available at http://zomax.wins.uva.nl:5345/ret_user/.

References [19].

Features Color and shape invariants are defined to be used as features in content based queries independent of camera viewpoint, geometry of the objects and illumination conditions. Under the assumption of a dichromatic reflection model, the effect of various imaging conditions on the different color features is analyzed. (In this model, reflection is used to determine how color clusters are formed in the RGB color space by pixels coming from the same uniformly colored surface of the object.) Assuming a white illumination and a matte object surface, a basic set of irreducible color invariants is formed by $\{R/G, B/R, G/B\}$. Also, it is proven that the normalized color model rgb and a newly defined model $c_4c_5c_6$, with $c_4 = (R-G)/(R+B)$, $c_5 = (R-B)/(R+B)$ and $c_6 = (G-B)/(G+B)$, are independent of the viewpoint, illumination direction, illumination intensity and surface orientation of matte objects. The set of color invariants, independent of viewpoint, used in the case of a shiny surfaced object, are $l_4 = |R-G|/(|R-G| + |B-R| + |G-B|)$, $l_5 = |R-B|/(|R-G| + |B-R| + |G-B|)$, $l_6 = |G-B|/(|R-G| + |B-R| + |G-B|)$. Based on the above color invariants, normalized color histograms are computed, H_A defined on the c_4, c_5 and c_6 axes and H_B on l_4, l_5, and l_6 axes. Color invariant gradients are defined using a multi-band approach and a histogram H_C of $l_4l_5l_6$ color edges is computed. That is because the color gradient for $l_4l_5l_6$ is the one that measures the presence of material edges (not shadow or highlight edges) in an image.

The shape invariants used are the angle between two color edges (invariant under rigid motion plus scaling) and the cross-ratio (projective invariant) between four color edges originating in the same point. A 1D histogram H_D is constructed on the discretized angle values, and a similar one H_E on the cross ratio values. Color and shape invariants are also combined in a 4D histogram H_F such that $H_F(i,j,k,l)$ counts the number of color edge maxima points of (i,j,k) color and generating an angle l.



Figure 3: PicToSeek result, query at left.

Querying The query image can be selected from the database collected by the system's Web crawlers or brought in by the user by giving a URL address, see figure 3. Before submitting the query, the user has to select the desired invariance, which indicates to the system how to choose the invariants in computing the similarity. Queries by user-specified feature values are also allowed.

Matching The similarity measure between a query histogram H_j^Q and a database image histogram H_j^I is the normalized intersection $d(H_j^Q, H_j^I) = (\sum_{k=1}^{n} min\{H_j^Q(k), H_j^I(k)\})/(\sum_{k=1}^{n} H_j^Q(k))$, with $j \in \{A, B, C, D, E, F\}$ and n is the number of nonzero bins in H_j^Q.

Result presentation The retrieved images are shown without explicit order.

3.5 SMURF

(Similarity-based Multimedia Retrieval Framework)

Developer Center for Geometry, Imaging, and Virtual Environments, Institute of Information and Computing Sciences, Utrecht University, The Netherlands.

URL `http://give-lab.cs.uu.nl`.

References [65].

Features It is a framework that incorporates color, texture, and shape, but the system is primarily oriented towards shape-based retrieval research. In particular there are demo's for polyline matching, and weighted point set matching.

Querying The search is started by selecting an image from a random selection. The user can select a new image from the search results so as to narrow down to the target images.



Figure 4: Hieroglyphic retrieval with SMURF.

Matching Matching of two polylines is done by computing the area between the two corresponding turning angle functions, and minimizing over rotations (vertical shift of the function), and starting position (horizontal shift). The weighted point set matching is done with a pseudometric, the Proportional Transportation Distance. Unlike the Earth Movers Distance, it obeys the triangle inequality for sets of unequal weight, which is essential for the indexing method used.

Indexing The triangle inequality and similarity of similarity measures is used to make the search more efficient. This is based on the following observation. Consider a shape or point set A_1 that closely matches a query A_q: $d(A_1, A_q)$ is small. Let A_r be some reference shape, called vantage object. If the triangle inequality holds, $d(A_r, A_q) \leq d(A_r, A_1) + d(A_1, A_q)$, then we know that $d(A_r, A_q) - d(A_r, A_1)$ is small as well. We can measure the distance between a database shape A_1 and a query A_q by comparing their distances from a vantage object A_r. The computation of the latter distances can be done off-line for all A_i. So at run-time the computation of a single $d(A_r, A_q)$ yields all A_i that resemble A_r as much as A_q, including those that resemble A_q. Using multiple vantage objects, say m, increases precision, recall is always 100%. A similar approach was taken in FIDS [5]. In SMURF, the search is done with an optimal approximate k-nearest neighbor search in the m-dimensional 'vantage space' of distances. In this way, the number of expensive similarity measures between shapes is reduced to m. The search further only takes cheap distance computations in vantage space.

Result presentation The results are shown in decreasing order of match score.

Applications The polyline matching is applied to about 70,000 polylines from a collection of hieroglyphics that is used as a font for a hieroglyphic text processor, see figure 4. The weighted point set matching is used for searching in a collection of 1100 fish contours.

3.6 SYNAPSE

(SYNtactic APpearance Search Engine)

Developer Center for Intelligent Information Retrieval, University of Massachusetts, Amherst, MA.

URL http://hobart.cs.umass.edu/~mmedia/. A demo of the system is available at the address http://cowarie.cs.umass.edu/~demo/Demo.html.

References [36].

Features Queries are made on the visual appearance of the objects in the image, defined as the shape of the intensity surface and represented using the outputs of a set of Gaussian derivative filters. More precisely, each image is filtered at uniformly sampled points with Gaussian derivative up to the order 2. Thus, for each sampled point $p = (x, y)$ of intensity $I(p)$, the so-called 2-jet at this point is given by the vector $J^2[I](p, \sigma) = \{I_\sigma(p), I_{x,\sigma}(p), I_{y,\sigma}(p), I_{x^2,\sigma}(p), I_{xy,\sigma}(p), I_{y^2,\sigma}(p)\}$, where $I_{x^i y^i,\sigma}(p)$ is the image convoluted with the (i, j)-th partial derivative of the Gaussian function of scale σ: $I_{x^i y^i,\sigma}(p) = I(p) * \sigma^{i+j} \partial^{i+j} G(p, \sigma)/\partial x^i \partial y^i$, $i, j \in \{0, 1, 2\}$. From this vector, four differential invariants are computed: $d_1^\sigma = I_x^2 + I_y^2$, $d_2^\sigma = I_{x^2} + I_{y^2}$, $d_3^\sigma = I_{x^2} I_x^2 + 2 I_{xy} I_x I_y + I_{y^2} I_y^2$, $d_4^\sigma = I_{x^2}^2 + 2 I_{xy}^2 + I_{y^2}^2$. Computing these four invariants at three different scales, $\sigma_1, \sigma_2, \sigma_3$, a multi-scale invariant vector D is generated for each sampled point p. Along with each multi-scale invariant vector D, the coordinates (x, y) of the corresponding sampled point and the number i of the image are also stored.

Querying Querying is done by example (see figure 5 where a region has been marked).

Matching Matching is done in two phases. First, for every invariant vector D_n of the query image, a number of invariant vectors stored in the database are found so that they lie in the range $[D_n - t, D_n + t]$.



Figure 5: Synapse query with the drawn region of interest.

This is done by finding for each component $D_n(j)$ a list of invariant vectors, whose j-component lies in the interval $[D_n(j) - t(j), D_n(j) + t(j)]$ and intersecting the lists.

In the second phase, a spatial check is performed on the list of candidate images. For each sampled point q_m of the query image, the number S_m of query points whose Euclidean distance to the q_m is within a threshold factor of the Euclidean distance of their corresponding points in the candidate image. The score of the candidate image is given by $\max_m S_m$.

Indexing An index is created on each of the 12 coordinates of the invariant vector D. Instead of storing each invariant vector in one record of the form (i, x, y, D), 12 smaller tables (inverted file lists), each with 4 columns $(D(p), i, x, y)$, are generated. Each inverted file list is stored in a binary tree sorted on the first column.

Result presentation Images are showed in decreasing similarity score.

Relevance feedback The search can be refined by marking the relevant images found among the results of a query and reiterate the query.

3.7 VisualSEEk

Developer Image and Advanced Television Lab, Columbia University, NY.

URL http://www.ctr.columbia.edu/VisualSEEk/.

References [55].

Features In the database population step, each image is automatically decomposed into regions of equally dominant colors. For each region, feature properties and spatial properties are retained for the subsequent queries. A query consists of finding the images that contain the most similar arrangements of similar regions.

The color region extraction uses the back-projection technique. The first step is the selection of a color set. This is a 166-dimensional binary vector c, which defines a selection of 166 colors in the HSV color space. From a given image I, another image B is generated by $B[x, y] = \max_j a[k, j]c[j]$, where $k \in \{0, \ldots, 165\}$ is the index of the color of the pixel (x, y) in the image I and $a[k, j]$ is the similarity between two colors (with the indices k and j) in the HSV space. Next the image B is filtered and spatially localized color regions are extracted.

Along with the color set used for back-projection, the region centroid, area (defined as the number of pixels in the region) and the width and height of the minimum bounding rectangle are also stored.

Querying To start a query, the user sketches a number of regions, positions and dimensions them on the grid (see figure 6) and selects a color for each region. Also, the user can indicate boundaries for location and size and/or spatial relationships between regions. After the system returns the thumbnail images of the best matches, the user is allowed to search by example using the returned images.

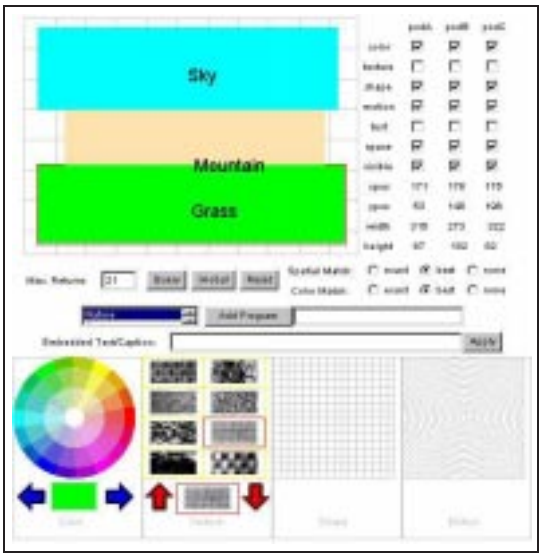


Figure 6: VisualSEEk query interface.

Matching To find the matches of a query image with a single region, queries on color set, region absolute location, area and spatial extent are first done independently. The color set similarity is computed by $d(c_q, c_t) = (c_q - c_t)^t A(c_q - c_t)$, where c_q, c_t are two color sets and $A = (a[i, j])$ is the color similarity matrix. If the user has defined spatial boundaries for the query region, then its distance to a target region is 0 if the target region centroid falls inside this boundaries, and is given by the Euclidean distance between the centroids otherwise. The distance in area between two regions is the absolute value of the difference, while the distance between the minimum bounding rectangles, (w_q, h_q) and (w_t, h_t), of two regions is the L_2 metric. The results of these queries are intersected and from the obtained candidate set, the best matching images are taken by minimizing a total distance given by the weighted sum of the four distances mentioned.

If the query image consists of a number of regions, in absolute or relative location, then for each region positioned in absolute location, a query like that described above is made, and for regions positioned by relative location individual queries on all attributes except location are performed. For the intersection of all this query results, the relative spatial relations specified by the user are evaluated using 2D string representation [11].

Indexing The color set distance can be written as $d(c_q, c_t) = \mu_q + \mu_t - 2c_q^t r_t$, where $\mu_q = c_q^t A c_q$, $\mu_t = c_t^t A c_t$ and $r_t = A c_t$. For each target region, μ_t and $r_t[m]$, $m \in \{0, \ldots, 165\}$, are indexed individually. The centroids of the image regions are indexed using a quadtree. For the indexing of the minimum bounding rectangles, R-tree are used.

Result presentation The results of a query are displayed in decreasing similarity order. Under each retrieved image, the distance to the query image is indicated.

3.8 WISE

(Wavelet Image Search Engine)

Developer Department of Computer Science, Stanford University.

URL http://www-db.stanford.edu/~wangz/project/imsearch/. A demo of the system is available at http://bergman.stanford.edu/~zwang/project/imsearch/WBIIS.html.

References [67].

Features The system, also called WBIIS (Wavelet-Based Image Indexing and Searching), performs queries on color layout information encoded using Daubechies wavelet transforms. In a preprocessing step, the image is rescaled to 128×128 pixels and converted from RGB color space representation to another color space. This color space, defined by $C_1 = (R + G + B)/3$, $C_2 = (R + (max - B))/2$, $C_3 = (R + 2 * (max - G) + B)/4$, with max the maximum possible value for each RGB color component, is selected because of its similarity to the opponent color axes used by the human visual system. On each of the three color components, a 4-layer 2D fast wavelet transform using Daubechies wavelets is applied. From the resulting 128×128 matrices, denoted as W_{C_1}, W_{C_2}, and W_{C_3}, each 16×16 upper left corner is stored as part of the feature vector (the 8×8 upper left corner, representing the lowest frequency bands in the wavelet transforms, accounts for object configurations in the image, while the three surrounding 8×8 submatrices represent detail information). Also, the standard deviation, σ_{C_i}, of the 8×8 upper left corner of W_{C_i} is computed and stored as part of the feature vector.

Querying Querying is done by example, where the query image is either a database image or a hand drawn sketch. Also low resolution images or partial images (images that contain non-specified areas represented by black rectangles) can be query images to the system.

Matching The matching process is performed in two steps. In the first step, the standard deviations computed for the query image, $\sigma^q_{C_i}$, are compared with the corresponding values, $\sigma^i_{C_i}$, stored for each database image. If the acceptance criterion, $(\sigma^q_{C_1}\beta < \sigma^i_{C_1} < \sigma^q_{C_1}/\beta) \vee ((\sigma^q_{C_2}\beta < \sigma^i_{C_2} < \sigma^q_{C_2}/\beta) \wedge (\sigma^q_{C_3}\beta < \sigma^i_{C_3} < \sigma^q_{C_3}/\beta))$, with the threshold β usually set to 0.5, is not achieved, the distance between the two images is set to 1. In the second step, the images that passed the first matching phase are compared to the query using a weighted Euclidean distance between wavelet transform coefficients: $\sum_{m=1}^{2}\sum_{n=1}^{2} w_{mn}\sum_{i=1}^{3}(w_{C_i} d_E(W^i_{C_i,m,n}, W^q_{C_i,m,n}))$, where $W_{C_i,1,1}$ and $W_{C_i,1,2}, W_{C_i,2,1}, W_{C_i,2,2}$ denotes the 8×8 upper left corner and, respectively, the three surrounding 8×8 submatrices of W_{C_i}, and with d_E the Euclidean distance. By raising w_{C_2} or w_{C_3} the importance of color variation in the image is emphasized, while raising $w_{2,1}, w_{1,2}$ or $w_{2,2}$ the vertical, horizontal, or diagonal edge details in the image are emphasized.

Result presentation The first few matches are presented in decreasing similarity order.

Applications The system is used at Stanford University Library to assist teaching and research projects in liberal art departments.



Figure 7: WISE result with top left query.

4 Summary

The features used in the 50 systems discussed in this article are listed in table 1. The features are classified into the classes color, texture, and shape, layout and face detection. The use of keywords is also listed.

We have tried to group the low level features into meaningful categories. For example, various forms of choosing important colors are grouped into 'dominant colors'. The 'eigen image' is listed under color, because it is a feature derived from the global image color values, rather than local patterns as in texture. The 'atomic texture features' are features such as contrast, anisotropy, density, randomness, directionality, softness, and uniformity, often variations of the Tamura features [59], and often derived from a cooccurrence matrix of pixel values. The 'edge statistics' include the edge image histogram, edge orientation histogram, and others. Recall that we consider edge orientation over all pixels as texture, but edge orientation at only region contours as shape information, see section 2. The 'elementary shape descriptors' are features such as centroid, area, major axis orientation, eccentricity, circularity, and features derived from algebraic moments. The feature 'layout' is the absolute or relative spatial position of the color, texture, or shape features. Taking color histograms per fixed subimage is a simple way of exploiting spatial layout; considering the relative position (constellation) of regions is more complex.

		ADL	AltaVista	Amore	ASSERT	BDLP	Blobworld	CANDID	C-bird	CBVQ	Chabot	CHROMA	DrawSearch	Excalibur	FIDS	FIR	FOCUS	ImageFinder	ImageMiner	ImageRETRO	ImageRover	ImageScape	Jacob	KIWI	LCPD	MARS	MetaSEEk	MIR	NETRA
	Reference	[68]	[1]	[40]	[50]	[8]	[9]	[27]	[33]	[54]	[43]	[30]	[48]	[16]	[5]	[66]	[14]	[24]	[28]	[64]	[49]	[31]	[10]	[34]	[32]	[44]	[4]	[57]	[35]
	Keywords	⋆		⋆		⋆					⋆											⋆					⋆	⋆	⋆	
	Face Detection																						⋆						⋆	
	Layout				⋆		⋆		⋆			⋆			⋆		⋆		⋆		⋆	⋆				⋆		⋆	⋆
Shape	No details available		⋆											⋆															
	Other								⋆[3]											⋆[5]									
	Elementary descriptors				⋆		⋆		⋆	⋆							⋆		⋆			⋆							
	Bounding box/ellipse																		⋆										
	Curvature scale space																												
	Elastic models																												
	Fourier descriptors												⋆									⋆				⋆			⋆
	Template matching			⋆																									
	Edge direction histogram																												
Texture	No details available		⋆											⋆				⋆											
	Other																								⋆[7]				
	Edge statistics				⋆				⋆						⋆						⋆		⋆						
	Local binary patterns														⋆							⋆			⋆				
	Random fields												⋆																
	Atomic texture features				⋆		⋆	⋆											⋆		⋆	⋆	⋆			⋆	⋆		
	Wavelet, Gabor, Fourier	⋆								⋆					⋆	⋆								⋆		⋆			⋆
Color	No details available		⋆											⋆				⋆											
	Other					⋆[1]			⋆[2]											⋆[4]		⋆[6]							
	Dominant colors			⋆				⋆	⋆			⋆							⋆			⋆							
	Region histogram				⋆		⋆																						⋆
	Color coherence vector																												
	Color moments				⋆								⋆							⋆				⋆					
	Correllation histogram																											⋆	
	Global/subimage histogram								⋆	⋆	⋆				⋆		⋆			⋆	⋆		⋆			⋆	⋆	⋆	
	Eigen image																												

Photobook	[45]	⋆											⋆									⋆							⋆	
Picasso	[6]						⋆															⋆		⋆	⋆			⋆		
PicHunter	[13]		⋆	⋆		⋆																								⋆
PicSOM	[29]				⋆												⋆[9]		⋆		⋆							⋆		
PicToSeek	[19]		⋆																							⋆[8]				
QBIC	[42]		⋆		⋆							⋆								⋆					⋆			⋆		⋆
Quicklook	[12]		⋆	⋆	⋆	⋆			⋆[10]		⋆	⋆			⋆				⋆						⋆			⋆		
RETIN	[18]							⋆			⋆																			
Shoebox	[38]				⋆				⋆[11]																					⋆
SIMBA	[51]																⋆[12]													
SMURF	[65]		⋆																							⋆[13]				
SQUID	[39]																						⋆							
Surfimage	[41]	⋆	⋆								⋆	⋆					⋆[14]		⋆			⋆								
SYNAPSE	[36]																⋆[15]													
TODAI	[37]														⋆															⋆
Viper	[56]		⋆					⋆			⋆																	⋆		
VIR	[2]									⋆								⋆									⋆			⋆
VisualSEEk	[55]							⋆																⋆	⋆			⋆		
VP	[26]																									⋆[16]				
WebSEEk	[53]		⋆																											⋆
WebSeer	[58]		⋆						⋆[17]																				⋆	⋆
WISE	[67]										⋆																			

1 number of regions with that color
2 DCT of chromaticity
3 generalized Hough transform
4 greyness, background proportion
5 number of regions, number of holes
6 Laplacian
7 projection histogram
8 angles between edges, and cross ratios of them
9 local probability
10 spatial chromaticity histogram
11 color channel variance
12 histogram of polynomial of local intensities
13 weighted point sets, polylines
14 intensity surface curvature histogram
15 multiscale intensity surface derivatives
16 convex parts
17 farthest neighbor histogram

Table 1: Overview of systems and their search features.

Of the 50 systems in the table, 41 use any kind of color features, 33 use texture, 26 use shape, 18 layout, and 4 use face detection. We conjecture that the reason for this order of frequencies of usage is their complexity and, coupled to that, effectiveness. The simpler the features can be extracted, the easier it is to implement it in the system, and the easier it is to use that feature class in the right way. For example, color features are often found effective, because good color features are not very difficult to design and implement. However, shape features that are for example robust against noise and occlusion, are more involved. As a result, only overly simple features such as area are used, which is often little effective.

5 Conclusions

Most systems are products of research, and therefore emphasize one aspect of content-based retrieval. Sometimes this is the sketching capability in the user interface, sometimes it is the new indexing data structure, etc. Some systems exist in a research version and in a commercial or production version. The commercial version is usually less advanced, and shows more standard searching capabilities. For example, a research version of Amore exhibits sketching and more fancy result visualization than is shown in the Arthur (Art Media and Text Hub and Retrieval) application system, http://www.isi.edu/cct/arthur/.

A number of systems provide a user interface that allows more powerful query formulation than is useful in the demo system. For example, if a user can paint a cartoon, but the database contains only airplanes, the system will always retrieve images of airplanes. For such a limited database, no powerful sketching interface is needed. Also, because most workstations have a mouse, but easy sketching needs a pencil, such a painting tool is often of little use. Drawing polygonal shapes with a mouse works well, however.

Most systems use color and texture features, fewer systems use shape feature, and still less use layout features. The retrieval on color usually yield images with similar colors. Retrieval on texture does not always yield images that have clearly the same texture, unless the database contains many images with a dominant texture. Searching on shape gives often surprising results. Apparently the shape features used for matching are not the most effective ones.

Indexing data structures are often not used. Indeed, for small collections of images, an indexing data structure is not needed, and a linear search can be sufficiently fast. Contemporary computers can perform simple matching of hundreds of images in near real time.

It is difficult to evaluate how successful content-based image retrieval systems are, in terms of effectiveness, efficiency, and flexibility. Of course there are the notions of precision (the ratio of relevant images to the total number of images retrieved) and recall (the percentage of relevant images among all possible relevant images). Many articles about systems give figures about precision and recall. Most of them are good, but hard to verify. One reason is that many hyperlinks on the Web are not active anymore, a design flaw of the Web. However, there are also considerations intrinsic to retrieval systems.

If the database only contains fighting airplanes, and the user can only ask for images similar to a chosen fighting airplanes, the system will successfully return fighting airplanes. If the domain is so narrow, it may make more sense to look for dissimilar images than for similar images. On the other hand, if the database is very diverse and contains only a single image of a chicken, asking for images similar to that chicken will not result in other chicken images. The larger the collection of images, the more chance that it contains an image similar to the query image. The Web is a large enough test set, and free of charge, reason why some image retrieval systems exploit webcrawlers. Having a specific object in mind, looking for images with similar objects is a frustrating experience, however. Indeed, before you learn to walk, you crawl first.

This survey shows that most current content-based image retrieval systems work with low level features (color, texture, shape), even the more experimental research systems. It is widely recognized that next generation systems should operate at a higher semantic level. One way to achieve this is to let the system recognize objects and scenes. Although this is difficult in general, it should be feasible for specific domains. For example, the ImageMiner system classifies landscapes, and body plans have been used to recognize animals and people [17]. Once separate entities in images are recognized, the way to semantic reasoning lies open.

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